

Do Prison reforms work: Evidence Based on the Justice Reinvestment Initiative*

Debbie Amewu[†]

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Abstract

Owing to the explosion in incarceration rates, coupled with high crime and overcrowded prisons in America, much research has been conducted to analyse the causes of such phenomena. This paper contributes to such research by conducting a policy analysis of the effectiveness of the Justice Reinvestment Initiative, a reformative legislation instituted to reform the justice system and tackle the issue. Since its inception, no empirical analysis has been conducted in the literature to ascertain its impact on crime rates. This paper employs the Difference-in-Differences estimator to review the impact the policy has on crime rates. The results show that the policy's implementation led to significant increases in property and total crime rates, contrary to popular belief in its successes.

Keywords: Justice Reinvestment, Prison Reform, Criminal Justice System

JEL Classification: K42, H76, K14

[†]Economics Department, University of Warwick, Coventry, United Kingdom

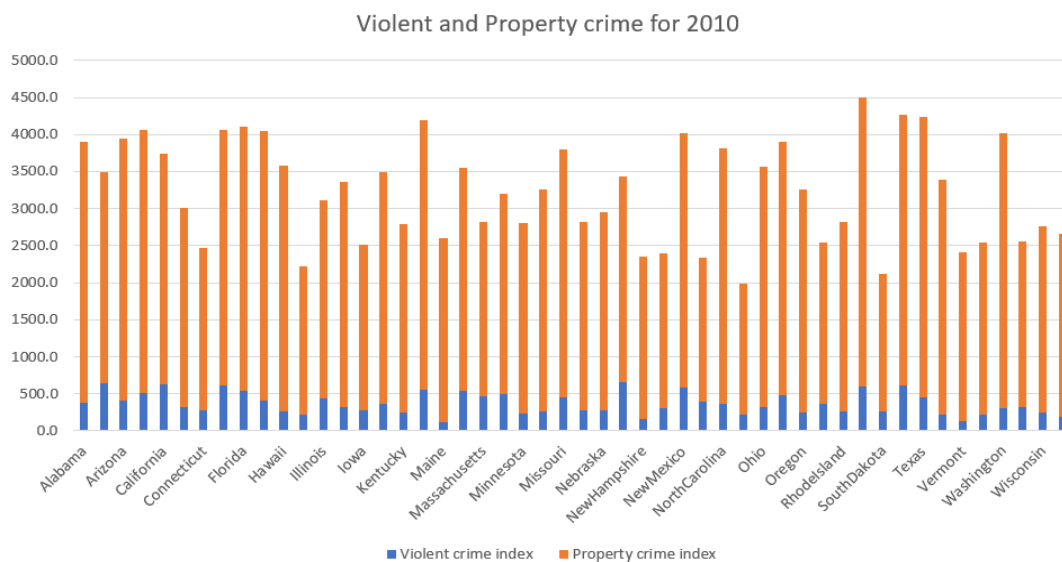
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1. Introduction

The United States has battled high crime and incarceration rates since the 1970s. Scholars believe that this unique phenomenon is due to the view held by justice system officials and legislators that incarceration is the main solution to tackling high crime rates and insecurity within society. Policies like the three strikes law, mandatory minimum, and the truth-in-sentencing led to indiscriminate incarceration and prison overpopulation. The resulting effect is that the justice system could no longer perform its primary function of reforming the individual and preventing re-offense. This, in turn, caused crime rates to consistently rise. The United States now has about a quarter of its population in prisons and jails (National Research Council, 2014).

However, starting from the early 2000s, crime rates in America have begun to fall. Nonetheless, the estimated number of reported property crime cases, for instance, was 5.2 million in 2018. Property crime is the most common form of crime in America. The graph below shows crime classifications in all states in 2010.

Figure 1: Graphical analysis of crime classifications for all 50 states in 2010.



Source: Federal Bureau of Statistics Uniform Crime Reports of Crime in the United States

From the graph it be noted that the property crime rates are higher than violent crime rates highlighting its impact on the economy.

The purpose of this research is to answer the question: how did the United States succeed in reducing its high crime rates inherited from the 1970s? This paper tries to ascertain whether prison reforms and the restructuring of the justice system were the driving factors in diminishing crime rates experienced in present years or whether some other factors might have influenced this. The main strategy of this paper is to analyse the impact of the Justice Reinvestment Initiative, which involves both private and public advocates, on three crime rate classifications, namely Total crime, Total violent crime, and Property crime. The scope of analysis is the entire 50 states of America, and the time frame is the years 2000 to 2018. Crime rates for the control and treatment states are compared before and after the Justice Reinvestment Initiative was instituted in 2010. The goal is to ascertain whether the Justice Reinvestment Initiative fulfils its intended goal of reducing crime by shifting focus from excess incarceration to identifying key drivers of crime and effectively mitigating them. The project would answer the question: did the Justice Reinvestment Initiative succeed in reducing crime rates by employing the Difference-in-Differences technique to analyse its policy effect after implementation?

Fagan, J. & Freeman, R. B. (1999) note the trade-off between crime and legal work. They emphasize that the youth who often lack the requisite skills to find gainful employment within society tend to resort to illegal forms of work to make a living. This is typically the case in disadvantaged states in the United States, such as Kentucky, Alabama, and Louisiana. The result is a depreciation in human capital of the vibrant youth, which can be debilitating for the economy. High crime rates also tend to affect housing values, increase the actual mobility rates of residents and reduce the connectedness of members of the community with their neighbourhood (Taylor, R. B., 1995). The economic disadvantages of high crime rates encourage the reformation of the justice system in order to tackle the issue. It also motivates research into the economics of crime to determine the factors that influence crime, the need to reform the justice system, and the usefulness of analysing the effectiveness of existing reforms.

This project is divided into seven sections. Section 2 is the institutional background of crime, incarceration, and justice reinvestment. Section 3 is the literature review

section, which points out similar and relevant work done by scholars on the topic of crime and prison reforms. Section 4 describes the data and the manipulations made to enhance its quality and form. Section 5 describes the methodology, identification strategy, and model specification. Section 6 is the results and discussion section, which contains the conclusion of the regressions and explains the economic implications useful for policy implementation. Section 7 is a brief analysis of the failures of the Justice Reinvestment Initiative and suggestions on the way forward in terms of restructuring the justice system while effectively dealing with America's high crime and incarceration rates. Section 8 is the appendix section containing all tables with the results of this paper.

2. Literature Review

There is a wealth of research done by scholars to find the right solution when it comes to solving the high crime, high incarceration problem of the United States. It comes as no surprise since the United States has the highest incarceration rate in the entire world. Scholars have brainstormed to ascertain the root causes of the large disparity between the United States' incarceration rate and that of the rest of the world. The economic benefits of crime reduction include maintaining housing values within communities, reducing the tax burden on citizens, reducing state and federal spending on incarceration systems, etc (Centre for American Progress, 2012). This section would analyse such relevant literature and provide comparisons between their main findings and the results of this paper.

2.1 Empirical Literature

2.1.1 Analysis of the Justice Reinvestment Initiative

Sabol and Baumann (2020) made a comprehensive analysis of the effectiveness of the Justice Reinvestment Initiative. They claim that the Justice Reinvestment Initiative shifted focus from its initial purpose, which they believed would have indeed had a higher impact in reducing crime in America, to a more political reformatory process

that targeted merely the state justice agencies' performance. The authors and earlier proponents of the initial focus of the Justice Reinvestment Initiative believed that the shift in focus led to the neglect of communities that were indeed affected by high incarceration and prison population growth. It also resulted in the abandonment of the goal of reinvesting cost savings into deprived states that needed capacity building the most. In conclusion, the Justice Reinvestment Initiative did not achieve its goals of redirecting parts of incarceration expenditure towards community and state development, of delegating the responsibility of monitoring offenders from the state level to local communities to cut costs, and of ensuring efficiency and demonstrating how the policy led to the reduction in prison populations. This is the main literature of this project. My research aims to buttress the points made by Sabol and Baumann (2020) by providing empirical evidence of the failures of the Justice Reinvestment Initiative.

2.1.2 Main drivers of crime

To address the mismatched negative relationship between incarceration and crime rates, Clear (1996) compared the crime lessening and crime-enhancing aspects of imprisonment. He concludes that factors that increase crime, such as replacement of co-offenders, intel on prison life by citizens, and social factors such as inequality and broken homes, are far more potent than factors that reduce crime, like imprisonment. This gives new insight into the main drivers of crime to enable officials to better tackle them.

2.1.3 The effects of high imprisonment rates on society

Clear (2008) points out the effects high imprisonment rates have on members of the community and the community as a whole. Some of these effects, he notes, are obstruction of social networks and friendships, which negatively affect the mental and physical well-being of the family unit of the convict. High incarceration also hampers the labour market outcomes and economic well-being of the convict and members of his/her family. Lastly, contrary to the knowledge and opinions shared by policymakers and authorities in the penal system of America, high incarceration increases the criminogenic propensity of members of the community. Clear (2008)

believes that reformatory activities such as re-entry programs, rehabilitation programs, and alternatives to incarceration do not cause a significant dent in solving the problem because they only target the margins while neglecting impoverished places within the United States.

2.1.4 The effect of prison populations on crime

The findings of Levitt, S (1996) are relevant in understanding the effects that massive prison populations have on crime through high incarceration rates. He conducts an instrumental variable analysis of the Prison Overcrowding Litigation implemented in 1965, which had unique implementation processes allowing the author to estimate the effect of each stage on crime rates. His research found estimates that are much higher than previous research conducted on this topic. He found that incarcerating an additional prisoner reduces the number of crimes by almost 15% per year.

2.2 Comparison with existing literature

This project provides empirical evidence to back the theoretical conclusions of Sabol and Baumann's (2020) research. The results show that the Justice Reinvestment Initiative rather increased crime, and this could be particularly due to the shift in focus of justice reinvestment earlier stated.

Similar to the Justice Reinvestment Initiative, Clear's approach to dealing with crime targets the main drivers of crime and seeks to find ways to mitigate them rather than embarking on intensified imprisonment. However, this paper proves that such measures adopted in the Justice Reinvestment Initiative were not successful in reducing crime rates.

The Justice Reinvestment Initiative did not result in a drastic reduction in imprisonment rates; hence, in accordance with Clear's (2008) on the drivers of criminogenic tendencies of members of the community, the Justice Reinvestment Initiative would also not result in a permanent solution to the vicious cycle of high crime and incarceration. The results of this paper provide a possible explanation for this, as crime rates were found to have increased throughout the implementation of the Justice Reinvestment Initiative.

My approach is significantly different from that of Levitt, S (1996), who uses an instrumental variable approach with the prison overcrowding litigation, while my research employs the Difference-in-Differences estimator for policy analysis. Although his research provides useful results, his scope of analysis is limited to the twelve states in which the litigation was implemented; hence, not much can be said about the other 38 states on which research was not conducted. The Difference-in-Differences estimation estimates the effect of the Justice Reinvestment Initiative for the control group [26 states] to get the missing counterfactual for the treated group [24 states].

2.3 Relevance of the research

Much of the analysis made on the impact and success of the Justice Reinvestment Initiative has been theoretical. Organisations dedicated to touting the achievements of the Justice Reinvestment Initiative, the Urban Institute, and the Council of State Governments, but none have conducted an empirical analysis to verify their claims, to my knowledge. This paper attempts a comprehensive empirical analysis to support the claims of Sabol and Baumann's (2020) about the failures of the policy and to buttress the points raised by Clear (2008) on rethinking criminal justice and penal policy. The analysis is performed on the entire United States to improve causal inference.

3. Institutional Background

3.1 History of mass incarceration and the hidden costs of incarceration

Mass incarceration founds its roots during the abolition of slavery in 1865. African Americans were disproportionately targeted with petty crimes and were heavily burdened with court costs and fines, which they were unable to pay. The resulting convict leasing was a masked form of slavery where the convicts were sold during a legal contract as laborers to the highest private bidder. The repression and control of

African Americans started the rising growth in incarceration rates, which surpassed the growth rate of the population by ten times at that time (Huff Post, 2016).

High crime rates in the 1970s led to excessive imprisonment for violent and even non-violent, minor crimes. Prison capacity has been stretched in many states, creating inhumane living conditions for the inmates. This results in negative psychological effects on prisoners, leading to erratic behaviour, anxiety, and post-traumatic stress disorder. Such effects increased the rate of re-offense, thereby causing a vicious cycle of incarceration and high crime rates.

The effects of high incarceration rates are innumerable. For one, the associated financial costs per year of managing incarceration have been on the rise since the 1970s in America. Yet again, the cost of incarceration per year exceeds the \$80 billion figure often touted by the Bureau of Justice Statistics (The Marshall Project, 2019). Wagner, P & Rabuy, B (2017) provide a detailed breakdown highlighting hidden costs such as policing costs, payment of employees at public correction agencies, costs of contracting private correctional facilities, the huge costs shouldered by family members of convicts, etc., bringing the final estimate to \$184 billion per year.

3.2 Evolution of justice reinvestment and the mechanism of the Justice Reinvestment Initiative

Justice reinvestment efforts began in the early 2000s in response to the conundrum of high costs of incarceration, rising prison populations, and high crime rates. It started as an effort to reform the justice system, which had very harsh and unnecessary sentencing policies leading to prison overcrowding and eventually not achieving its goal of ameliorating crime rates by reforming the convict. The main objectives of justice reinvestment were to reduce societal exacerbation caused by mass incarceration which forcibly uproots members of the society and releases them unable to fit back into the society; to improve the transition and synchronizing of members of the society and ex-convicts by providing them with educational, counselling and job opportunities; and to take into account the needs of offenders such as the basic social opportunities and necessities of life while also accounting

for the public safety risk they pose (Sabol & Baumann, 2020). Justice reinvestment facilitated the reinvestment of cost savings from the incarceration system from state to local communities to enable the local management and supervision of offenders. It also ensured the reinvestment of cost savings into local organisations in deprived areas within America that would provide facilities such as counselling, educational, and financial support to offenders to reduce the rate of recidivism. These initiatives tend to reduce prison population as incarceration drops, which helps to reduce crime through reductions in the criminogenic tendencies of convicts. It also reduces crime rates through community development.

With time, the Justice Reinvestment Initiative became the dominant form of justice reinvestment in the United States. It adopted some of the previous measures but with a few changes. Instead of being a community-focused investment, it focuses on justice system agencies. Instead of focusing on reducing incarceration rates, it resorted to improving the efficiency of the justice system operations as a means of scaling down the costs of incarceration. Lastly, it gave prominence to state-level agency stakeholders as opposed to community advocates. All in all, the Justice Reinvestment Initiative continued to adopt the goals of reducing expenditures on incarceration, which would better the society and possibly reduce crime.

The Justice Reinvestment Initiative uses data collected by states and cities on the main drivers of crime and prison populations and associated costs to deduce more efficient ways to solve the issue of crime, reinvesting any cost savings into other policies that help to develop the communities' capacity to cope with offenders at the local level. The Justice Reinvestment Initiative was initiated in 2010 with a membership of 24¹ states. It is a bipartisan, public-private partnership endeavour between the Bureau of Justice Assistance and the Pew Charitable Trusts, which ensures efficiency in public safety measures by enacting legislative reforms that address corrections and sentencing norms that are inefficient and costly, replacing them with evidence-based practices that yield desired results.

¹ This number is as at 2017

3.3 Structure of the United States prison system

Early forms of punishment through imprisonment began in the 1500s, when people were held in detention establishments like dungeons. Since then, the American penal system has grown and evolved into the large body it is in present times. The American corrections system is a conglomerate of systems and agencies that are governed by laws, policies, and constitutional regulations. This conglomerate includes 50 separate state systems, a federal system, and many county corrections agencies. Each state has a certain level of monopoly to control its sentencing practices, although they are allowed to derive exemplification from federal sentencing guidelines (SAGE Publications, 2019). The majority of prosecutions are held at the state level, although some crimes, such as drug abuse, terrorism, or federal tax fraud, which are considered federal offenses, are handled at the federal level. Offenders who commit a crime against the state are held in state correctional facilities, while federal offenders are held in federal correctional facilities. The federal involvement in criminal justice legislation aims to ensure a sense of uniformity for state criminal justice codes. It also ensures that constitutionally guaranteed rights for all persons are not denied by the criminal justice system. (Friends Committee on National Legislation, 2020). The graph below shows the racial composition of federal prisons in America according to the Federal Bureau of Prisons.

Figure 2: Tabular representation of Inmate racial composition in Federal prisons in September 2020

	Race	# of Inmates	% of Inmates
	Asian	2,340	1.5%
	Black	59,679	38.4%
	Native American	3,638	2.3%
	White	89,627	57.7%

Source: Federal Bureau of Investigation

Per the illustration, federal prisons consist of 38.4% Black Americans, but Black Americans form only 13.4% of the American population (United States Census Bureau, 2019).

The United States correctional system makes use of jails, prisons, and solitary confinement to hold inmates. There are different security levels of prisons to hold prisoners based on the severity of the offense. Community supervision is also a frequently used method in incarceration and consists of probation and parole (Correctional Officer, 2020).

4. Data

4.1 Data sources

This study employs eight datasets. These include: total crime, total violent crime, total property crime from the Federal Bureau of Investigation Uniform Crime Reports of Crime in the United States; Unemployment rates from the Bureau of Labour Statistics' State Employment and Unemployment Archived News Releases; Population density² from the United States Census Bureau's State Intercensal Tables (2000-2010) and State Population Totals and Components of Change (2010-2019); the number of people in Gangs³ from the National Gang Centre's National Youth Gang Survey Analysis; High school dropouts from the National Centre for Education Statistics' Trends in High School Dropout and Completion Rates in the United States: 1972-2012 Compendium Report; Recidivism rates from the Virginia Department of Corrections' Three year re-incarceration rates Reports⁴; Various programming/reformative services available at state correctional facilities and the

² The values from the United States Census Bureau were divided by the size of the state in square feet to obtain Population density values.

³ Gross gang membership was totalled per each year from 2000 to 2012. This was manually divided by the population of each state to apportion gang membership for each state.

⁴ The rates of recidivism for each state are the earliest information available on re-offence rates over a three-year period

inmate to correctional officer ratio from the Bureau of Justice Statistics' database on such information; and Real GDP from the Department of Numbers. All variables are at the state level for the years 2000 to 2018. The variables for the Ordinary Least Squares are at the state level for the year 2005 because the Bureau of Justice Statistics only had recent information on programming services in the 2005 prison census. Therefore, all the other variables were matched with that particular time frame [2005].

4.2 Description of Variables

My research conducts two regression analyses: a Difference-in-Differences regression and an Ordinary Least Squares regression. The Difference-in-Differences regression analyses the policy effect of the Justice Reinvestment Initiative on crime rates in America after its implementation in 2010. The period of reference is the years 2000-2018, while the subject matter is all 50 states of America. The Difference-in-Differences estimator would analyse the treatment effect on the treatment group, which would not have occurred had the treatment not been implemented. My research also uses the Ordinary Least Squares regression method to ascertain the association between reformative program services offered in correctional facilities to rehabilitate convicts and recidivism rates in all 50 states of America. Due to the nature of the data for reformative programs services, which is only available for 2005, the linear regression is conducted for the year 2005 for the reformative programs services whiles the recidivism is a three-year average rate occurring as of 2018. Hence, the strategy is to ascertain the relationship between reformative program services offered in the past [2005] and recidivism rates occurring in the present [in 2018]. Below are descriptions of the outcome variables and covariates for the Differences-in-Differences regression and the Ordinary Least Squares regression. All variables in both regressions are logged.

4.2.1 Description of Outcome Variables for the Difference-in-Differences

The outcome variables for the Difference-in-Differences estimation are the total crime index, the total violent crime index, and the total property crime index. The indices are calculated per 100,000 citizens for each crime classification for the years 2000 to 2018. According to the Federal Bureau of Investigation, total violent crimes are offenses that impose force or threat of force on the victim. Total violent crime constitutes murder and nonnegligent manslaughter, forcible rape, robbery, and aggravated assault. My research would use the aggregate figure, and not the individual categories, for violent crime in each state from 2000 to 2018. The Federal Bureau of Investigation defines property crime as theft-type offenses, which do not include force or threat of force, but displace the victim of his/her personal belongings in the form of money or property. Property crime comprises burglary, larceny-theft, motor vehicle theft, and arson⁵. The total crime index is the sum of the total violent index and the total property crime index. The Federal Bureau of Investigation Uniform Crime Reports uses a Hierarchy Rule, which takes into account the most serious offense that is committed during a multiple-felony scenario and reports only that. This applies to both total violent crime and total property crime.

4.2.2 Description of Outcome Variables for the Ordinary Least Squares

The outcome variables for the Ordinary Least Squares regression are the three-year recidivism rate in 2018. Recidivism is the instance where an ex-offender recommit a crime after being released or during his period of parole or probation.

4.2.3 Description of Control Variables for the Difference-in-Differences

My Difference-in-Differences estimation uses the following controls: unemployment rate, population density, number of people in gangs, and high school dropout rate.

⁵ Arson incidences are not included in the totals although they are considered in the trend, clearance, and arrest tables because, according to the Federal Bureau of Investigation Uniform Crime Reports, data are not available. This is because of different data collection procedures by law enforcement agencies and limited participation in such exercises.

The unemployment rates are the end-of-year seasonally adjusted rates collected for each state for the years 2000 to 2018.

For population density, this paper combines two unique datasets from the United States Census Bureau, as previously mentioned in the data sources section. The first dataset provides population estimates from the year 2000 to 2009. Such estimates are generated by modifying the postcensal estimates from April 1, 2000, to identify any differences between then and the 2010 census counts. The modification process entails updating the resident population from the 2000 census by the estimates of the components⁶ of population change between 2000 and 2018. The second database provides population estimates from 2010-2018. For the second dataset, the United States Census Bureau employed a cohort-component method derived from the demographic balancing equation:

Figure 3: Graphical representation of the methodology for population estimates for the United States.



Source: United States Census Bureau. METHODOLOGY FOR THE UNITED STATES POPULATION ESTIMATES: VINTAGE 2019 Nation, States, Counties, and Puerto Rico – April 1, 2010 to July 1, 2019

The estimate is calculated by adding births, subtracting deaths, and adding net migration, which is both international and domestic, to the population base (last decennial census or previous point in the time series). This paper then conducted data manipulation to arrive at the population density by dividing the population estimates per state and year by the size of each state in square miles.

The total number of people in gangs per state estimates was compiled from the national youth survey analysis, which reports the total estimated number of gang members from 1996 to 2012. These figures were apportioned on a state basis using

⁶ These components include; births to United States resident women, deaths to United States residents, domestic migration, international migration, and net movement of the United States armed forces.

population estimates derived from the United States Census Bureau that have already been described. The time frame for this variable is 2000 to 2012, since the National Gang Centre only had available information up to the year 2012.

High school rates are collected from the NCES, who measure State event dropout rates for public school students in grades 9-12. The event dropout rate refers to the percentage of high school students who left school between the beginning of one school year and the beginning of another school year without earning a high school diploma or alternative credential like the General Education Development (GED). This project uses high school dropout rates from 2000 to 2012 due to limitations of the dataset.

4.2.4 Description of Covariates and Explanatory variables for the Ordinary Least Squares regression

For the Linear regression, there are five explanatory variables, namely Inmates provided with work programs, Inmates participating in work release, Facilities Providing counselling, Facilities providing secondary education, Inmate per correctional officer. These were collected at the state level for the year 2005. The covariates used in this regression are similar to those of the Difference-in-Difference regression except for the inclusion of Real GDP and the elimination of Population Density. These are also collected at the state level for 2005.

4.3 Justification for covariates

4.3.1 Justification of controls for Difference-in-Differences regression

There are many factors that influence crime in America apart from prison reforms. Common examples are level of education, real income level, employment rates, etc. To not bias the estimates, covariates are added to increase confidence in the estimates derived from the Linear Regression. The controls for my Difference-in-Differences regression are unemployment rates, the number of people in gangs, population density and high school dropout rates. Unemployment rates were added to account for its influence on crime rates. Crime is negatively related to legal work hence, the higher unemployment rates are, the higher the tendency for involvement

in illegal work (Fagan, J. & Freeman, R. B., 1999). Population density varies positively with crime. The higher the population growth in a particular state, the less the economic opportunities are available forcing inhabitants to resort to vices like theft and burglary to make ends meet. Watts, E R (1931) notes that the influence of population density on crime is not limited to the aggregate growth rate of population alone but it is necessary to recognise the gender and age group which have the highest offense rate. In this regard, the use of mere population density only gives a snapshot into the influence of population on crime rates. Gang membership increases crime rates through its psychological impact on the individual. Rampant high school dropouts could lead teenagers to resort to illegal activities to make ends meet. According to the United States Department of Commerce, Census Bureau, Current Population Survey, March 2013, the median income of individuals who did not complete their high school education or equivalent earn \$21,000 less than their counterparts who receive a high school diploma or GED certificate. This could serve as a window of opportunity for high school dropouts who earn less to indulge in criminal activities to bridge the income gap.

4.3.2 Justification of covariates for Ordinary Least Squares regression

Cornaglia, F. et al (2014) note the negative impact of violent crime on the mental wellbeing of convicts. This could prove a link to trends in recidivism rates as longer periods of imprisonment impacts the rate of re-offense. This explains the relevance of the use of Recidivism as the dependent variable in the regression. The explanatory variables used in this paper are reformatory programs provided at the correctional facilities on the state level. The services include providing work programs, work release, counselling, and secondary education. Providing occupational activities and work release to inmates provides the relevant skills help them fit into the community after release. Counselling services provides therapeutic solutions to inmates shaping them to become better individuals within the society. Secondary school education equips inmates with the required skills to find jobs once they complete their sentences. These factors tend to reduce the likelihood of ex-convicts to re-offend. The inmate to correctional officer ratio provides a glimpse of the quality of security

services and reformative services given per each inmate in the state correctional facilities. The more officers there are, the better the facility can manage its inmates. According to Clear (1996), the three main crime enhancers in the United States are the replacement of co-offenders, intel on prison life and social-economic factors. The inclusion of the covariates controls for these factors. Replacement of co-offenders is the instance where members of a gang initiate a new member when one of the existing members gets arrested by the police. Clear (1996) believes that this increases the propensity of offense of the most unlikely within the society. This also leads to recidivism through the affinity convicted members of the gang have to the group. The number of people in gangs per state is added to the regression to reflect this [replacement of co-offenders]. The higher the number of gang members, the higher the rate of recidivism. Unemployment rate, high school drop out rate and real GDP control for socio-economic factors. Below are summary statistics of the variables for both regressions.

Table 1: Summary statistics of Treatment states.

Dependent variables	Mean	Std. Deviation	Minimum	Maximum	Observations
Total violent crime rate per 100,000	378	154.2	103.5	885.0	570
Property crime rate per 100,000	3017	780.6	1248.5	5781.7	570
Total crime rate per 100,000	3395	867.8	1421.7	6043.7	570
Explanatory variables	Mean	Std. Deviation	Minimum	Maximum	Observations
Unemployment rate	6	2.0	2.0	14.5	570
Number of people in gangs	7241	8525.5	0	34579.3	570
Population density	181	240.9	1.1	1039.4	570
High school drop-out rate	3	2.2	0	9.2	570
Other variables⁷	Mean	Std. Deviation	Minimum	Maximum	Observations
Murder and nonnegligent murder	4.7	2.2	1.1	10.8	24
Forcible rape	33.1	14.1	20.3	79.7	24
Robbery	84.8	40.1	15.2	172.3	24
Aggravated assault	241.0	96.5	108.4	433.2	24
Burglary	699.2	223.8	391.0	1081.3	24
Larceny theft	2081.9	317.8	1447.7	2588.3	24
Motor vehicle theft	195.0	64.0	77.5	303.1	24

⁷ The values correspond to the year 2012

Table 2: Summary statistics for control states

Dependent variables	Mean	Std. Deviation	Minimum	Maximum	Observations
Total violent crime rate per 100,000	393.5	171.9	77.8	856.6	380
Property crime rate per 100,000	2935.4	878.8	1283.1	5833.4	380
Total crime rate per 100,000	3328.9	996.1	1455.1	6386.3	380
Explanatory variables	Mean	Std. Deviation	Minimum	Maximum	Observations
Unemployment rate	5.6	2.1	2.0	14.9	380
Number of people in gangs	14514.7	20183.7	0	95717.7	380
Population density	207.6	282.1	5.1	1208.3	380
High school drop-out rate	2.5	2.4	0	11.5	380
Other variables⁸	Mean	Std. Deviation	Minimum	Maximum	Observations
Murder and nonnegligent murder	3.8	1.5	1.3	7	26
Forcible rape	28.8	8.5	11.7	46.4	26
Robbery	87.6	46.1	10.6	178.3	26
Aggravated assault	224.6	97.2	61.0	479.6	26
Burglary	597.4	187.3	329.9	1025.3	26
Larceny theft	1865.7	311.1	1,383.7	2439.1	26
Motor vehicle theft	201.8	91.9	69.5	443.2	26

Table 3: Summary statistics for all states combined for 2005 (Ordinary Least Squares regression)

Dependent Variable	Mean	Std. Deviation	Minimum	Maximum	Observations
Recidivism	38.1	10.5	21.4	64.5	50
Explanatory Variable	Mean	Std. Deviation	Minimum	Maximum	Observations
Inmates provided with work programs	15506.6	22092.2	28	123419	50
Inmates participating in work release	25682.6	33845.8	1411	169988	50
Facilities providing counselling	34.3	29.2	6	132	50
Facilities providing secondary education	25.9	22.5	4	102	50
Inmate per correctional officer	4.7	1.5	0	9	50
Covariate⁹	Mean	Std. Deviation	Minimum	Maximum	Observations
Number of people in gangs	15758.6	17298	1373.6	95717.7	50
Unemployment rate	4.8	1.1	2.7	8.8	50
High school dropout	3.7	1.7	0	8	50
Real GDP	46361.7	8171.4	30813	67525	50

⁸ The values correspond to the year 2012⁹ These statistics are based on the year 2005 and differ from those provided under the Difference-in-Differences statistics because they pertain to the years 2000 to 2018. Also the recidivism rate is the three-year average as at 2018.

5. Methodology

This paper focuses on the policy analysis of the Justice Reinvestment Initiative and its effects on crime rates through prison population and incarceration. Using cross sectional or time series estimators poses some challenges. The cross-sectional estimator might be faced with omitted variables that might cause variations between the treatment and control groups and bias the estimates. The time series estimator might neglect factors that affect the variable of interest in both control and treatment groups differently over time. The Difference-in-Differences method resolves the issue of unobserved heterogeneity that cross-sectional and time-series estimators are unable to.

5.1 Identification Strategy

5.1.1 Identification Strategy for Difference in Differences

The Difference-in-Difference estimator is used to estimate the causal impact of a policy implemented for two unique groups namely the treatment and the control group. In this paper, the treatment group comprises the states that were affected by the policy implementation while the control group comprises the states that were not affected by the policy implementation. The Difference-in-Differences estimator is used to ascertain the average treatment effect on the treated [the treatment group]. The key goal is to find a control group that represents the missing counterfactual that would have existed if the treatment group were not treated. This control group must be similar in all characteristics to the treatment group, in this case, in terms of state-specific characteristics of the United States, but the only difference is that those states did not receive the treatment. The Difference-in-Differences estimator then estimates the treatment effect which is the difference between the impact on the treatment group and the counterfactual treatment group [control group] (Joshua D. Angrist, 2009). Another key feature of the estimator is that the selection of the treatment and control states in this project should be as good as random. In this case, the eligibility of the state to receive the treatment [Justice Reinvestment legislations] is based on the completion of the Justice Reinvestment Initiative model which is an

application process to vet the state's suitability for the Initiative. Therefore, states cannot self-select into the treatment group. Selection is based on the ability of the state fulfil the requirements enlisted in the application process. Some of these requirements include the ability to form a bipartisan, interbranch task force or committee, to analyse criminal justice data to determine the factors causing the increase in crime, to implement evidence-based strategies etc. This ensures that selection bias is kept at a minimum.

My research involves the use of the Difference-in-Difference estimator on the entire state population of the United States to derive a comprehensive impact of the Justice Reinvestment Initiative on crime rates in America. Hence, the treatment group¹⁰ is those states that participated in the Justice Reinvestment Initiative from 2010 to 2018. The control states¹¹ are all the other states in the United States that did not participate in Justice Reinvestment Initiative. The time frame remains the years 2000 to 2018. Since all states belong to a single country, the structure and response to economic phenomenon are quite similar. Also, all treatment states are concentrated in the South Eastern portion of the United States while the controls are in the Western parts of the United States. Crime rate trends in the control and treatment groups are also similar. There are both high crime and low crime states in the control and treatment groups.

5.1.2 Identification Strategy for Ordinary Least Squares regression

The second identification strategy involves the use of an Ordinary Least square regression analysis to ascertain the correlation between recidivism rates over a three-year period as at 2018 for each state and various programming services offered in all state correctional facilities in 2005. The linear regression process involves ascertaining the correlation between programming services offered in 2005 and recidivism rates occurring after 2005. The sample size remains the 50 states of

¹⁰ These 24 states include New Hampshire, South Carolina, Arkansas, Kentucky, Louisiana, North Carolina, Ohio, Delaware, Georgia, Hawaii, Mississippi, Missouri, Oklahoma, Pennsylvania, Kansas, Oregon, South Dakota, West Virginia, Idaho, Alabama, Nebraska, Utah, Alaska, and Maryland.

¹¹ These include New York, Arizona, California, Colorado, Connecticut, Florida, Illinois, Indiana, Iowa, Maine, Michigan, Minnesota, Nevada, New Jersey, New Mexico, Tennessee, Texas, Vermont, Virginia, Washington, Wisconsin, Wyoming, Massachusetts, Montana, North Dakota, and Rhode Island.

America whiles controls namely Gang membership, high school dropout rates, real GDP and unemployment rates for the year 2005 are included.

5.2 Model Specification

5.2.1 Model Specification for Difference-in-Differences

The estimating equation is:

$$T_{st} = \beta_0 + \beta_1\gamma_t + \beta_2J_s + \beta_3(\gamma_t * J_s) + \eta X_{st} + \mu_{st}$$

$$V_{st} = \beta_0 + \beta_1\gamma_t + \beta_2J_s + \beta_3(\gamma_t * J_s) + \eta X_{st} + \mu_{st}$$

$$P_{st} = \beta_0 + \beta_1\gamma_t + \beta_2J_s + \beta_3(\gamma_t * J_s) + \eta X_{st} + \mu_{st}$$

$$s = 1, 2, \dots, 50 \quad \& \quad t = 1, 2, \dots, 19$$

The three equations represent the Difference-in-Differences estimation of the impact of the Justice Reinvestment Initiative on the three classifications of crime where T_{st} represents Total crime index per state (s) at time (t), V_{st} represents Total violent crime index per state (s) at time (t) and P_{st} represents Total Property crime index per state (s) at time (t). γ_t and J_s are binary indicators for time and treatment status respectively. The binary indicator for time represents the periods before and after 2010. Where,

$$\gamma_t = 0 \text{ if before 2010}$$

$$\gamma_t = 1 \text{ if after and during 2010.}$$

The binary indicator for treatment status tells whether the state was treated or not. Where,

$$J_s = 0 \text{ if state was treated}$$

$$J_s = 1 \text{ if state was not treated}$$

Lastly, X_{st} is a vector of covariates namely unemployment rate per state (s) at time (t), gang membership per state (s) at time (t), population density per state (s) at time (t), and high school drop-out rate per state (s) at time (t).

The outcomes to be observed for average effects for the treatment and control groups are found in Table A7 in the appendix.

Table A6 in the appendix shows the conditional mean estimates of the Difference-in-Differences estimator. β_3 , which is the main result of the Difference-in-Differences estimator, represents the difference between the difference in average crime rate for the treatment and control group in pre-treatment and the difference in average crime rate for the treatment and control group in years following post-treatment (Roberts, M. R., n.d.). β_1 is the average crime rate of treatment states in the pre-treatment era relative to the average crime rate of control firms in the pre-treatment era [across all three crime classifications]. β_2 is the average crime rate of control firms in the post-treatment era relative to the average crime rate of control firms in the pre-treatment era. $\beta_1 + \beta_3$ is the average crime rate in the treatment group relative to the control group in the post treatment era. The Difference-in-Differences estimator differences away any permanent and unobservable differences between the control and treatment groups and any other common trend affecting both groups.

The key foundation of the Difference-in-Differences estimator is the concept of parallel trends. The parallel trends assumption implies that in the absence of the treatment, the treatment and control groups would have evolved similarly in the outcome variable. This means that if the treatment had no altering effect on the outcome variables of the control and treatment group post-treatment then the average treatment effect and the average treatment effect on the treated would be equal.

5.2.2 Model Specification for the Ordinary Least Squares regression

The estimating equation is found below:

$$R_s = \alpha_0 + \alpha_1 W_s + \alpha_2 WR_s + \alpha_3 C_s + \alpha_4 S_s + \tau_5 \rho_s + \varepsilon_s$$

$$s = 1, 2, \dots, 50$$

Where R_s is the three-year Recidivism rate per state as at 2018. W_s, WR_s, C_s, S_s, I_s are the programming services offered namely work programs, work release, counselling services, secondary education and the inmate per correctional officer respectively.

G_s , U_s , H_s are previously described as Gang membership, Unemployment rate and high school dropout rate but for the year 2005 solely. RG_s is Real GDP per capita for 2005.

Although the properties of the Ordinary Least Squares estimator are unbiasedness, consistency, efficiency and asymptomatic distribution, the results of this regression do not imply causation unless the expected value of the error term is zero (. This means that there is strictly no relationship between the explanatory variables and unobserved variables. In reality, it is almost impossible to ensure this hence, results from linear regressions are correlation or association results rather than causation results. The coefficients or the regressors represent the conditional mean of associations between the explanatory variables and the dependent variables.

6. Results and Discussion

6.1 Exploratory analysis

6.1.1 Correlation between recidivism and reformatory service at state correctional facilities.

To motivate the essence of my research, I began by performing an Ordinary Least Squares regression of recidivism rates over a three-year period and prison system reforms at correctional facilities at the state level per each state. These reforms include the number of inmates provided with work programs, the number of inmates provided with work release, the number of facilities providing counselling services, the number of facilities providing secondary education and the inmate per correctional officer ratio which is used as a measure of the quality of reformatory and security services the inmates receive while in prison. The hypothesis is that states with the highest quality of reforms at their jails and prisons would essentially have low re-offense or recidivism rates, all things being equal. The purpose of the Ordinary Least Squares regression is not to imply causality and provide a basis of

economic inference but rather to provide information about the associations between the variables I have chosen in this project. My results differ from the work of Mitchell, O et al (2016) who analyse the impact of imprisonment on recidivism in Florida by using the regression discontinuity design. The authors make conclusions that the alternative methods of incarceration would significantly reduce the likelihood of convicts to re-offend, especially those who commit minor non-violent crimes. In the same vein, my research concludes that reformatory services provided at incarceration centres are associated with reducing recidivism rates. Although my results are exploratory, my research broadens the scope of analysis to all United States rather than a sample of convicts within the America as done in Mitchell, O et al (2016).

The explanatory variables for this linear regression are five and recidivism rates were separately regressed on these explanatory variables sequentially. The coefficients of concern are specifically those from column 5 where all controls such as the number of people in gangs, unemployment rate, high school dropout rate and real GDP are added. The results from Tables A1-A5, in the appendix, mainly show negative correlations between all the reforms and recidivism rates except inmates participating in work release. In terms of significance, 'Facilities providing secondary education' and Facilities providing counselling are significant at the 5% level in models 5 and 1¹² respectively. The other two variables are not significant in any of the models therefore counselling and secondary education services offered at correctional facilities are associated with reductions in recidivism. This could be due to the strict boundaries preventing ex-convicts from re-entering the labour market due to the felony status. In conclusion, a one percent increase in Facilities providing secondary education is associated with a 0.15 reduction in recidivism rates while a one percent increase in Facilities providing counselling is associated with a 0.10 reduction in recidivism rates. This is crucial for policy analysis in that these results provide empirical evidence to the importance of investing into reformatory programs in correctional facilities where education and counselling services tend to

¹² Facilities providing counselling became insignificant when the covariates were added.

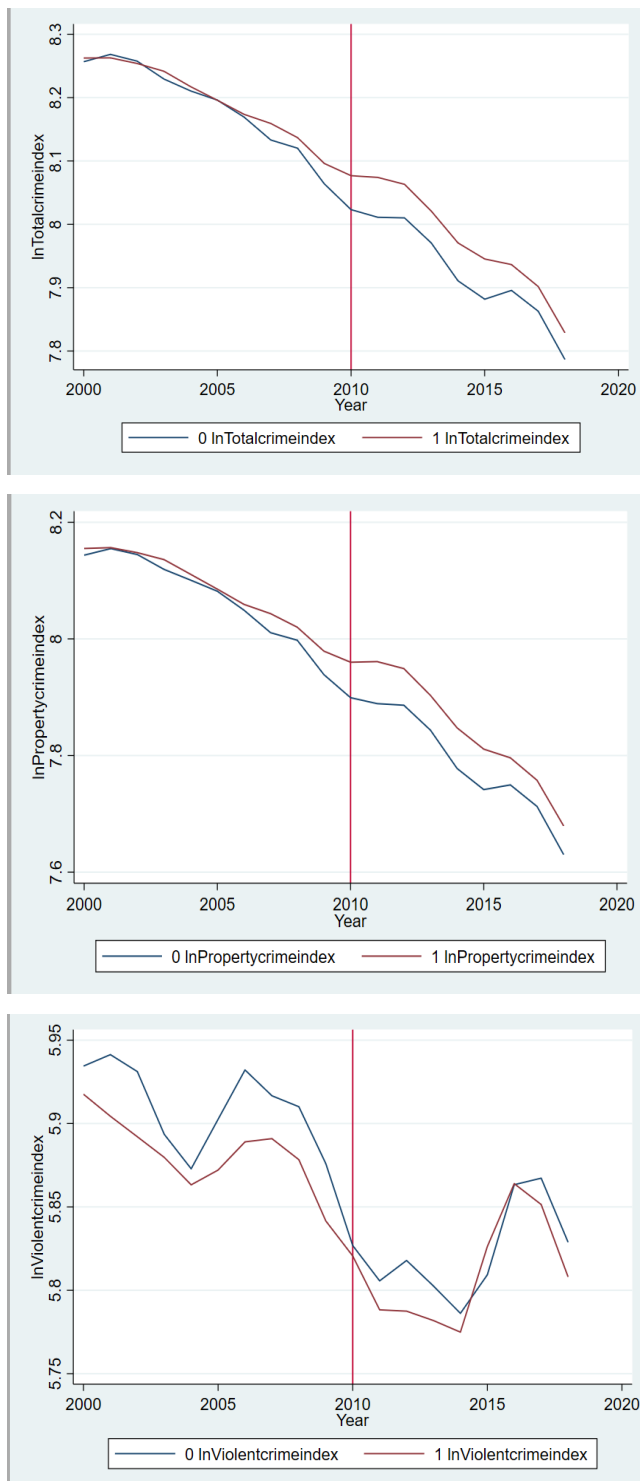
recidivism rates. Also, community capacity building should increase the accessibility of such educational facilities to improve standards of living in deprived areas of America. All the results of the correlation are found in Tables A7-A11.

6.2 Confirmatory analysis

6.2.1 Visualizing the parallel trend assumption within the states

A key assumption in implementing Difference-in-Differences estimation is the parallel trends assumption which implies that in absence of the treatment, the outcome variables in both the treatment and control groups would have evolved similarly. My dataset on each dependent variable is relevant enough to show this assumption in all classifications of crime. Per the results, the impact of the Justice Reinvestment Initiative on total crime and total property crime was positive whiles that of total violent crime was negative hence, the trend showed that Total crime rate and total property crime rate for the control states fall below that of the treatment states which is in line with the results whiles total violent crime rate for the control states lies above that of the treatment states. Below are figures for the trends that reflect this.

Figure 4: Graphical representation of the parallel trend within the states for all three crime classifications



6.2.2 Results from the Difference-in-Differences analysis.

The results from this analysis are found in Tables 4-9. The main results of the impact of the Justice Reinvestment Initiative legislation on the various classifications of

crime prior to the addition of the controls to the regressions are found in Tables 4-6 while the results with the inclusion of the controls are found in Tables 7-9. The tables below show the results of the Differences-in-Differences estimator using the three main dependent variables.

Table 4: Dependent variable: Total crime index

	Post Justice Reinvestment Initiative legislation	Pre-Justice Reinvestment Initiative legislation	Difference
Treatment (24 participating states)	$\beta_0 + \beta_1 + \beta_2 + \beta_3 = 7.980$	$\beta_0 + \beta_1 = 8.120$	$\beta_2 + \beta_3 = -0.220$
Control (Rest of the States)	$\beta_0 + \beta_2 = 7.928$	$\beta_0 = 8.191$	$\beta_2 = -0.262$
Difference	$\beta_1 + \beta_3 = 0.052$	$\beta_1 = 0.010$	$\beta_3 = 0.042$

Table 5: Dependent variable: Total violent crime index

	Post Justice Reinvestment Initiative legislation	Pre Justice Reinvestment Initiative legislation	Difference
Treatment (24 participating states)	$\beta_0 + \beta_1 + \beta_2 + \beta_3 = 5.811$	$\beta_0 + \beta_1 = 5.883$	$\beta_2 + \beta_3 = -0.071$
Control (Rest of the States)	$\beta_0 + \beta_2 = 5.823$	$\beta_0 = 5.911$	$\beta_2 = -0.088$
Difference	$\beta_1 + \beta_3 = -0.0112$	$\beta_1 = -0.028$	$\beta_3 = 0.017$

Table 6: Dependent variable: Property crime index

	Post Justice Reinvestment Initiative legislation	Pre Justice Reinvestment Initiative legislation	Difference
Treatment (24 participating states)	$\beta_0 + \beta_1 + \beta_2 + \beta_3 = 7.852$	$\beta_0 + \beta_1 = 8.089$	$\beta_2 + \beta_3 = -0.238$
Control (Rest of the States)	$\beta_0 + \beta_2 = 7.792$	$\beta_0 = 8.074$	$\beta_2 = -0.282$
Difference	$\beta_1 + \beta_3 = 0.059$	$\beta_1 = 0.015$	$\beta_3 = 0.044$

For all crime classifications, namely total crime, property crime and total violent crime, the Justice Reinvestment Initiative had a positive impact. The legislation of the Justice Reinvestment Initiative increased total crime, total violent crime, and property crime by 0.042, 0.017 and 0.044 crime per 100,000 people, respectively. However, these results are not statistically significant. With the addition of the controls, the estimates improve and become significant. As a robustness check, controls are added one after the other to monitor the magnitude and direction of the

estimates. For total crime and total property crime, the significance of the estimates increased with each addition of controls. However, the estimates of total violent crime did not gain significance with the addition of the controls. Both total crime and property crime had significant and positive coefficients of 0.05 and 0.054, respectively. On the other hand, the Justice Reinvestment Initiative had a negative impact on total violent crime of 0.001 per 100,000. Tables 7-9 below provide results of this analysis.

Table 7: Impact of the Justice Reinvestment Initiative Legislation on total crime

<i>Dependent variable is log total crime index</i>					
	(1)	(2)	(3)	(4)	(5)
Constant	8.191*** (0.06)	8.050*** (0.060)	8.518*** (0.20)	8.435*** (0.20)	8.25*** (0.19)
Justice Reinvestment Initiative legislation (Treatment)	0.010 (0.01)	0.007 (0.07)	0.001 (0.08)	0.008 (0.08)	0.008 (0.07)
Time	-0.262*** (0.02)	-0.271*** (0.02)	-0.140*** (0.15)	-0.141*** (0.02)	-0.133*** (0.02)
Justice Reinvestment Initiative legislation (Treatment) x Time	0.042 (0.03)	0.047 (0.03)	0.044* (0.02)	0.045* (0.02)	0.050** (0.02)
Log of Unemployment rate		0.088*** (0.02)	-0.086*** (0.01)	-0.084*** (0.01)	-0.072*** (0.01)
Log of Gang membership			-0.020 (0.02)	0.006 (0.02)	0.012 (0.02)
Log of Population Density				-0.037* (0.02)	-0.030 (0.02)
Log of High school dropout rate					0.058* (0.03)
Observations	650	650	650	650	612
Number of states	50	50	50	50	50

Notes: This table contains regression results on total crime index for all fifty states. Controls were added to achieve significant results. Robust standard errors are in parenthesis and coefficients are significant at the following confidence intervals:

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Table 8: Impact of the Justice Reinvestment Initiative Legislation on total violent crime

<i>Dependent variable is log total violent crime index</i>					
	(1)	(2)	(3)	(4)	(5)
Constant	5.911*** (0.12)	5.991*** (0.11)	5.401*** (0.40)	5.459*** (0.39)	5.151*** (0.45)
Justice Reinvestment Initiative legislation (Treatment)	-0.282 (0.15)	-0.027 (0.15)	0.007 (0.14)	0.001 (0.14)	2.072 (7.88)
Time	-0.088 (0.06)	-0.083 (0.06)	-0.067 (0.05)	-0.066 (0.06)	-0.050 (0.04)
Justice Reinvestment Initiative legislation (Treatment) x Time	0.017 (0.07)	0.014 (0.02)	0.001 (0.06)	-2.39e-06 (0.06)	-0.001 (0.00)
Log of Unemployment rate		0.050*** (0.02)	-0.060*** (0.02)	-0.060*** (0.02)	-0.037** (0.02)
Log of Gang membership			0.064* (0.04)	0.045 (0.05)	0.062 (0.05)
Log of Population Density				0.027 (0.05)	-0.025 (0.06)
Log of High school dropout rate					0.084* (0.05)
Observations	650	650	650	650	612
Number of states	50	50	50	50	50

Notes: This table contains regression results on total crime index for all fifty states. Controls were added to achieve significant results. Robust standard errors are in parenthesis and coefficients are significant at the following confidence intervals:

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Table 9: Impact of the Justice Reinvestment Initiative Legislation on property crime

<i>Dependent variable is log Property crime index</i>					
	(1)	(2)	(3)	(4)	(5)
Constant	8.075*** (0.06)	7.905*** (0.06)	8.484*** (0.20)	8.386*** (0.20)	8.220*** (0.18)
Justice Reinvestment Initiative legislation (Treatment)	0.015 (0.07)	0.013 (0.07)	0.003 (0.08)	0.011 (0.08)	0.010 (0.07)
Time	-0.282*** (0.02)	-0.292*** (0.02)	-0.146*** (0.01)	-0.147*** (0.01)	-0.139*** (0.02)
Justice Reinvestment Initiative legislation (Treatment) x Time	0.044 (0.03)	0.049* (0.03)	0.047** (0.02)	0.048** (0.02)	0.054*** (0.02)
Log of Unemployment rate		0.105*** (0.02)	-0.090*** (0.01)	-0.086*** (0.01)	-0.076*** (0.02)
Log of Gang membership			-0.028 (0.02)	0.002 (0.02)	0.001 (0.02)
Log of Population Density				-0.043** (0.02)	-0.036* (0.02)
Log of High school dropout rate					0.054* (0.03)
State controls	Yes	Yes	Yes	Yes	Yes
Observations	650	650	650	650	612
Number of states	50	50	50	50	50

Notes: This table contains regression results on total crime index for all fifty states. Controls were added to achieve significant results. Robust standard errors are in parenthesis and coefficients are significant at the following confidence intervals:

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

My results are in line with the conclusions of Sabol and Baumann's (2020) who believed that the Justice Reinvestment Initiative failed in its goal of reducing crime rates in America by reducing prison populations, channelling expenditure on the incarceration into other ventures which ensure community development and improving public safety. The authors believe that the deviation in the initial goals of justice reinvestment which included decentralising supervision of offenders to the local level and increasing accountability of offenders to their local communities, led

to a detrimental impact on crime rates. This occurred because the Justice Reinvestment Initiative failed to demonstrate practical ways to implement the procedures of reinvestment of funds. For instance, it failed to recognise that significant reductions in incarceration rates is the main source of reduction in monies spent on the American system that could be invested in more profitable areas of the economy (Sabol and Baumann, 2020). They also failed to ensure that monies saved would be invested to cater for the needs of the community rather than catering for state-level budgets (Sabol and Baumann, 2020). These failures are reflected in my results as the Justice Reinvestment Initiative led to an increase in property crime and total crime rates.

The by-product of the Justice Reinvestment Initiative is to reduce incarceration rates and prison populations by diverting attention from mass incarceration to community-based development. This makes the concept of this regression similar to Chambers, B., Das, S & Murali, A. (2013) who run regressions of incarceration rates on burglary and robbery rates in America. The main aim was to determine whether incarceration rates have the potential to increase burglary and robbery rates rather than reducing them as generally believed by legislative officials. My research extends analysis from violent crime to property crime and combines both classifications into total crime which were not covered in Chambers, B., Das, S & Murali, A. (2013). This increases the scope of causality in the regression results. Finally, the Differences-in-Differences estimator provides higher econometric power as compared to Ordinary Least Squares. It deals with omitted variables or unobserved heterogeneity among the states that would not be possible with the research of Chambers, B., Das, S. & Murali, A. (2013)

7. Conclusion

This paper has attempted to analyse the evolution of crime and incarceration rates in America. It began by illustrating the 1970s period of high crime rates and made attempts to explain why it happened within that period. This paper then proceeds to explain possible reasons why crime rates in the early 2000s began to drop across all states in America. It did so by employing various economic tools such as the Difference-in-Differences estimator and Ordinary Least Squares regression to try to explain the question of how crime rates and recidivism suddenly began to fall in America starting from the early 2000s. In doing so, this paper estimated the impact on crime caused by the Justice Reinvestment Initiative which was legislated in America in 2010 to reform the justice system. The results from that regression are statistically significant for total crime index and property crime index but for violent crime index, the impact was negative and statistically not significant.

Clear (2008) prompts the rethinking of crime policy as a solution. He believes that the two pressure points increasing incarceration and crime within America are the number of people who are imprisoned and how long they are imprisoned. With this in view, it is imperative that legislative bodies ensure the reduction in imprisonment and length of sentences. Sentencing reforms that would enable this include abandoning prison penalties for common drug crimes, mandatory minimum sentences for other felonies and technical revocations of probation and parole. This would go a long way to solve both epidemics of high crime and high incarceration in America.

My results are limited in scope because they do not provide an absolute reason for the reduction in crime rates in America. A single policy was analysed but there could have been other policies and legislations that could have equally impacted crime rates but were not included in the regressions due to lack of data. For instance, other policies like the Sentencing Reform Act were also implemented prior to the implementation of the Justice Reinvestment Initiative which could have impacted my results. Tonry, M (1984) pointed out the defects of the Sentencing Reform Act caused by political manipulation and deviation from purposed goals. This could

have accounted for increases in crime rates that the Differences-in-Differences regression could not pick up. Also, although the overall impact of the Justice Reinvestment Initiative was an increase in crime rates, the effect is not that large which might be indicating counteractive influences from other reformatory policies that are currently in operation at the state level. Lastly, the covariates introduced into the regressions are not the only factors that impact crime rates in recidivism within the economy. The neglect of such factors such as divorce rate per state, number of single parented families and drug abuse statistics could also have an impact on crime but were not added to reduce the volume of the results to be attained.

This paper is a novel empirical analysis of the impact of the Justice Reinvestment Initiative on crime rates in American and provides interesting revelations that are useful in reforming the justice system in America. The general notion is that these policies are always successful in accomplishing their targets, but sometimes empirical analysis could prove otherwise. This paper does just that.

8. References

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9. Appendix

8.1 Results from the linear regression with recidivism

Table A1: Relationship between total crime and inmates provided with work programs

Dependent variable is recidivism rate					
	(1)	(2)	(3)	(4)	(5)
Constant	3.924*** (0.23)	4.332*** (0.42)	4.105*** (0.427)	3.904*** (0.43)	-0.764 (3.29)
Log of Inmates provided with work programs	-0.037 (0.26)	0.001 (0.022)	-0.013 (0.03)	-0.020 (0.03)	-0.022 (0.03)
Log of Gang membership		-0.080 (0.06)	-0.082 (0.05)	-0.062 (0.06)	-0.072 (0.05)
Log of Unemployment rate			0.232 (0.20)	0.139 (0.18)	0.233 (0.17)
Log of High school dropout rate				0.173 (0.10)	0.061 (0.13)
Log of Real GDP					0.447 (0.31)
Observations	50	50	50	46	46
Number of states	50	50	50	50	50

Notes: The table reports the regression on recidivism rates. The explanatory variable is inmates provided with work programs and covariates are added to improve the estimates. Errors are clustered at the state level and coefficients are significant at the following levels:

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Table A2: Relationship between recidivism and inmates participating in work release

Dependent variable is recidivism rate					
	(1)	(2)	(3)	(4)	(5)
Constant	4.139*** (0.35)	4.378*** (0.41)	4.204*** (0.42)	3.962*** (0.42)	-0.646 (3.25)
Log of Inmates participating in work release	-0.056 (0.04)	0.087 (0.11)	0.059 (0.11)	-0.018 (0.13)	0.003 (0.12)
Log of Gang membership		-0.174 (0.13)	-0.158 (0.13)	-0.065 (0.14)	-0.100 (0.12)
Log of Unemployment rate			0.189 (0.20)	0.120 (0.18)	0.203 (0.16)
Log of High school dropout rate				0.170 (0.11)	0.051 (0.14)
Log of Real GDP					0.444 (0.31)
Observations	50	50	50	46	46
Number of states	50	50	50	50	50

Notes: The table reports the regression on recidivism rates. The explanatory variable is inmates participating in work release and covariates are added to improve the estimates. Errors are clustered at the state level.

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Table A3: Relationship between recidivism and Facilities providing counselling

Dependent variable is recidivism rate					
	(1)	(2)	(3)	(4)	(5)
Constant	3.93*** (0.15)	4.093*** (0.57)	3.868*** (0.52)	3.501*** (0.53)	-0.929 (3.36)
Log of Facilities providing counselling	-0.102** (0.05)	-0.073 (0.10)	-0.083 (0.09)	-0.129 (0.08)	-0.123 (0.08)
Log of Gang membership		-0.028 (0.09)	-0.039 (0.09)	0.007 (0.08)	-0.010 (0.07)
Log of Unemployment rate			0.229 (0.09)	0.128 (0.17)	0.214 (0.16)
Log of High school dropout rate				0.198* (0.11)	0.089 (0.13)
Log of Real GDP					0.427 (0.32)
Observations	50	50	50	46	46
Number of states	50	50	50	50	50

Notes: The table reports the regression on recidivism rates. The explanatory variable is facilities providing counselling and covariates are added to improve the estimates. Errors are clustered at the state level.

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Table A4: Relationship between recidivism and Facilities providing secondary education

	Dependent variable is recidivism rate				
	(1)	(2)	(3)	(4)	(5)
Constant	3.91*** (0.13)	3.969*** (0.56)	3.666*** (0.51)	3.325*** (0.51)	-0.998 (3.33)
Log of Facilities providing secondary education	-0.109** (0.04)	-0.102 (0.08)	-0.124* (0.07)	-0.157** (0.07)	-0.150** (0.07)
Log of Gang membership		-0.008 (0.08)	-0.012 (0.07)	0.022 (0.07)	0.006 (0.06)
Log of Unemployment rate			0.266 (0.18)	0.172 (0.17)	0.254 (0.16)
Log of High school dropout rate				0.203* (0.11)	0.096 (0.13)
Log of Real GDP					0.417 (0.32)
Observations	50	50	50	46	46
Number of states	50	50	50	50	50

Notes: The table reports the regression on recidivism rates. The explanatory variable is facilities providing secondary education and covariates are added to improve the estimates. Errors are clustered at the state level.

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

Table A5: Relationship between recidivism and inmate per correctional officer

	Dependent variable is recidivism rate				
	(1)	(2)	(3)	(4)	(5)
Constant	3.820*** (0.19)	4.683*** (0.43)	4.508*** (0.44)	4.353*** (0.43)	1.369 (3.58)
Log of inmate per correctional officer	-0.147 (0.12)	-0.106 (0.12)	-0.129 (0.12)	-0.142 (0.12)	-0.093 (0.14)
Log of Gang membership		-0.101** (0.04)	-0.121*** (0.04)	-0.120*** (0.04)	-0.116*** (0.04)
Log of Unemployment rate			0.253 (0.17)	0.150 (0.16)	0.199 (0.15)
Log of High school dropout rate				0.178* (0.10)	0.108 (0.13)
Log of Real GDP					0.279 (0.34)
Observations	50	50	50	46	46
Number of states	50	50	50	50	50

Notes: The table reports the regression on recidivism rates. The explanatory variable is inmate per correctional officer and covariates are added to improve the estimates. Errors are clustered at the state level.

***Significant at the 1% level

**Significant at the 5% level

*Significant at the 10% level

8.2 Conditional mean estimates and average outcomes

Table A6: Conditional Mean Estimates

	Post 2010	Pre 2010	Difference
Treatment group	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	$\beta_0 + \beta_1$	$\beta_2 + \beta_3$
Control group	$\beta_0 + \beta_2$	β_0	β_2
Difference	$\beta_1 + \beta_3$	β_1	β_3

Table A7: Average outcomes for treatment and control groups in the Difference-in-Difference

Estimator		
¹³	Treatment group	Control group
Before 2010	$E(C_{it} \mid J=1, \gamma=0)$	$E(C_{it} \mid J=0, \gamma=0)$
During and After 2010	$E(C_{it} \mid J=1, \gamma=1)$	$E(C_{it} \mid J=0, \gamma=1)$

¹³ Notes: J and γ represent dummy variables that describe the status of the treatment and time respectively. The resulting outcome, crime rate, is represented by C and the reference period is 2010 [t=1 is during and after the reference period and t=0 is before the reference period].

